MILITARY SPECIFICATION

POWER UNIT ASSEMBLIES: 8289242 AND 8619600

1. SCOPE

1.1 This specification covers two types of electro-mechanical units known as Power Unit Assemblies, 8289242 and 8619600, used individually on Periscope Mount M115 to illuminate the reticle of the Periscope M31 (T50) and forms a part of the fire control system for the 105MM Gun Tank M60.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

UU-T-111	Tape, Paper Gummed (Sealing and Securing)
PPP-T-76	Tape, Pressure-Sensitive Adhesive, Paper Water Resistant
PPP-B-585	Boxes, Wood, Wire Bound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock Corner
PPP-B-636	Boxes, Fiber
PPP-C-843	Cushioning Material, Cellulosic

Military

MIL-B-131	Barrier Material, Water-Vapor-proof, Flexible
MIL-P-116	Preservation; Methods of
MIL-P-17667	Paper, Wrapping, Chemically Neutral
MIL-F-13926	Fire Control Material; General Specification Governing the Manufacture and Inspection of
MIL-F-14619	Fire Control Parts, Equipment and Tools; Packaging of

FSC 1240

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

DRAWINGS

Ordnance Corps

D8289242	Power Unit Assembly
D8619600	Power Unit Assembly
LSQAP 8289242	Supplemental Quality Assurance Provisions List
LSQAP 8619600	Supplemental Quality Assurance Provisions List

(Copies of specifications, standards, drawings and publications required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production models.- At the beginning of production one power unit, representative of production, shall be made available to the Government to determine if the power unit meets all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Construction.- The power units shall be manufactured in accordance with Drawing D8289242 for Power Unit 8289242 and Drawing D8619600 for Power Unit 8619600, and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

3.4 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Protection finished surfaces

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(c) Special accuracy
(d) Finish painting and protective
(e) Workmanship

3.5 Performance

3.5.1 Contact function.- The contact (eyelet) shall move freely from its fully extended position approximately 7/64 inches when applying a force in the direction of movement. Spring action shall be such that when applied force is removed contact shall return to its original position.

3.5.2 Rheostat knob torque.- The amount of torque required to rotate the rheostat knob shall not exceed 9.6 ounce-inches nor be less than 3.6 ounce-inches at room temperature (60° to 90°F).

3.6 Electrical check.

3.6.1 Continuity.- The electrical circuits shall be tested for continuity and switch function and shall conform to the electrical schematic.

3.6.2 Power unit function.- With the circuit energized from a 24 ± 0.5 volt source, rotation of the rheostat knob shall vary the voltage from zero to 24 ± 0.5 volts.

3.7 Environmental.

3.7.1 Operating temperatures.- The power unit shall meet all the performance requirements of this specification while exposed and thermally stabilized at ambient temperatures of plus 150 degrees Fahrenheit and minus 40 degrees Fahrenheit.

3.7.2 Storage temperatures.- The power unit shall show no evidence of physical failure and shall meet all performance requirements of this specification at room temperature (plus 60 to 90 degrees Fahrenheit) after having been exposed and thermally stabilized at ambient temperatures of plus 160 degrees and minus 80 degrees Fahrenheit.

3.7.3 Vibration.- With the power unit mounted in normal operating position, it shall be vibrated in a vertical plane at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 minutes plus or minus 15 seconds. Subsequent to vibration there shall be no evidence of loose parts or damage and the power unit shall meet all performance requirements of this specification.

3.7.4 Rheostat knob torque.- The amount of torque required to rotate the rheostat knob shall not exceed 10.6 ounce-inches at temperatures of minus 40 to plus 150 degrees Fahrenheit.

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3.7.5 Water resistance.- The power unit shall meet the requirements of Procedure I, rain test of Specification MIL-F-13926.

3.8 Identification marking.- Identification marking unless otherwise specified shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection.- The contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality assurance system.- The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering all components and assemblies of this specification. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when as a minimum it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.
(b) Contractor's quality standards of acceptability.
(c) Procedure for continually evaluating product quality.
(d) Procedure for continually reporting evidence of actual product quality.
(e) Measures for preventing any significant departure of product quality from standards.

4.1.1.1 The contractor will not be restricted to the inspection station or to the method of inspection listed in the written procedure provided an equivalent control is included in the approval quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure of applicable drawings and specifications shall be used. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that might affect the degree of assurance required by this specification and the other documents referenced herein.

4.1.2 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of:

(a) Surveillance of the operations to determine whether the practices, methods and procedures of the written inspection plan are being properly applied.

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(b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until corrections have been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances.

4.2 Production model.

4.2.1 Unless otherwise specified in the contract, one unit taken from the production run shall be tested as specified herein by the contractor under the surveillance of the Government.

4.2.2 The test shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment designed or procured for use on the remaining items.

4.2.3 Production model test failure.- Should the production model fail to comply with any of the requirements specified, the model shall be rejected. Acceptance of production units shall cease until objective evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency.

4.3 Inspection provisions.

4.3.1 Inspection of components.- Inspection requirements for component parts and subassemblies are contained herein and on Supplementary Quality Assurance Provisions (SQAP) listed on Supplementary Quality Assurance Provisions List LSQAP 8289242 for the Power Unit 8289242 and Supplementary Quality Assurance Provisions List LSQAP 8619600 for the Power Unit 8619600.

4.3.1.1 The contents of SQAP shall represent the minimum inspection requirements and shall be the basis for acceptance or rejection by lot of those components covered by a SQAP.

4.3.1.2 Contractor's inspection for all components not covered by SQAP shall be in accordance with the inspection procedures of the written inspection plan required by 4.1.1. The plan shall include procedure for checking all positional and functional requirements that have a bearing on the functional accuracy and performance of the power units.

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4.3.1.3 Control of materials.- Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right in any case to require retests to assure that the materials are acceptable.

4.3.2 Lot formation.- Unless otherwise specified by the contracting officer or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.3.3 Acceptance tests.- The characteristics as listed in Tables I and II shall constitute the minimum inspection requirements of the power unit to be performed by the contractor prior to acceptance or rejection by lot. The Government reserves the right, however, to inspect for any applicable requirement and to reject individual non-conforming items.

4.3.3.1 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE I - CLASSIFICATION OF DEFECTS

EXAMINATION INSPECTION

Use Inspection Level II and Sampling Plan Table IVA of Standard MIL-STD-105

AQL value is expressed in percent defective

CHARACTERISTIC TO BE INSPECTED MAJOR: AQL 1.0	Req't Par.	Method of Inspection
Performance		
101. Contact function	3.5.1 & 3.7.2	ME
102. Rheostat knob torque	3.5.2 & 3.7.2	STE
103. Electrical continuity	3.6.1 & 3.7.2	ME
104. Voltage	3.6.2 & 3.7.2	ME
Preservation, Packaging and Packing		
105. Cleaning of unit prior to packaging	5.1.1.1	Visual
106. Wrapping improper or inadequate	5.1.1.4	Visual
107. Cushioning improper or inadequate	5.1.1.4	Visual
108. Container not to spec.	5.1.1.4	ME - Visual
109. Intermediate container not to spec.	5.1.1.5	ME - Visual
110. Closing and strapping improper or inadequate (Level A & B)	5.2.1 5.2.2	Visual
111. Marking improper or inadequate	5.3	Visual

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TABLE II - INSPECTION TESTING

Use Inspection Level L7 and Sampling Plan Table IVA of MIL-STD-105

AQL value is expressed in percent defective

CHARACTERISTIC TO BE TESTED AQL 6.5	Req't Par.	Test Method or Procedure
1. Rheostat knob torque (+150°F to -40°F)	3.7.1 & 3.7.4	STE
2. Voltage and sequence (+150°F to -40°F)	3.7.1 & 3.6.2	ME
3. Extreme temperature storage (+150°F to -40°F)	3.7.2	Visual & STE
4. Vibration	3.7.3	Visual & STE
5. Water resistance	3.7.5	MIL-F-13926

4.3.4 Inspection equipment.

4.3.4.1 Contractor provisions.- Unless otherwise specified in the contract, and contractor shall provide and properly utilize all gaging, measuring and testing equipment of the required accuracy for the performance of all inspection examinations and tests required to be performed by the contractor under this contract. The contractor shall be responsible for the maintenance of the equipment including surveillance, calibration and rehabilitation on a regularly scheduled basis, and shall have available all calibration equipment and other instrumentation essential for calibration and certification of the inspection equipment. The contractor also shall maintain records of inspection equipment surveillance and calibration and shall make these records available for review by the Government. The Government reserves the right to certify inspection equipment and to use it for Government verification inspection or acceptance inspection.

4.3.4.2 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes, established by product requirements, shall be capable of measuring these values within an accuracy of ten percent of the specified tolerance. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.4 Test methods and procedures.

4.4.1 Initial tests.- The units selected shall be submitted to the following tests after which they shall be submitted to the remaining tests of this specification:

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(a) Extreme temperature storage tests.- The units shall be placed in separate chambers for the extreme temperatures specified in 3.7.2. They shall be so placed that there is no obstruction to the flow of air across and around them. Direct heat conduction to the specimens should be minimized. Each specimen shall be exposed to one temperature extreme for a period of two hours, removed from the chamber, allowed to return to room temperature, then placed in a chamber at the other temperature extreme for a repeat of the cycle mentioned. Upon completion of these cycles, the units shall meet the requirements of 3.7.2.

(b) Vibration tests.- The units shall be mounted, vibrated and examined in accordance with 3.7.3.

4.4.2 Tests at -40°F and +150°F.- The units selected shall be placed in a suitable chamber and the necessary external controls attached. After stabilization as specified in 3.7.1, the units shall be submitted to the following tests:

(a) Voltage and sequence.- a 24 ± 0.5 volt source that has been checked with a voltmeter shall be used for the supply voltage in this test. When the contact and the supply voltage are connected as specified in 3.6.1, a voltmeter (20,000 ohms per volt sensitivity, and plus or minus 2% of full scale accuracy) shall be utilized. Voltage shall vary from 0 at the "on" position to 24 ± 0.5 volts at the point of maximum clockwise rotation.

(b) Rheostat knob torque.- During voltage and sequence tests the torque required to move the rheostat shall be measured with equipment specified in 4.3.4.2 and shall meet the requirements of 3.7.4.

4.4.3 Continuity.- The electrical circuit shall be checked for continuity by use of standard measuring equipment and shall conform to the wiring diagram specified in 3.6.1.

4.4.4 Water resistance test shall be conducted in accordance with Procedure I of Specification MIL-F-13926.

4.4.5 Knob torque at room temperature.- With the units stabilized at room temperature, torque testing equipment shall be attached and the rheostat rotated from the "off" position to the extreme clockwise position while noting the torque required. The requirements of 3.5.2 shall be satisfied.

4.4.6 Contact function.- The contact (eyelet) shall be checked for travel by applying a force and measuring the distance traveled as specified in 3.5.1.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified (See 6.1).

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5.1.1 Level A.

5.1.1.1 Cleaning.- The power unit assemblies shall be thoroughly cleaned in accordance with process C-1 (any applicable process) of Specification MIL-P-116.

5.1.1.2 Drying.- The power unit assemblies shall be dried in accordance with one or more of the drying procedures listed in Specification MIL-P-116.

5.1.1.3 Preservative application.- Not applicable.

5.1.1.4 Unit packaging.- The unit protection shall be Method IA-15 of Specification MIL-P-116. Immediately after drying, the power unit shall be wrapped with chemically neutral paper conforming to Type II, Class I of Specification MIL-P-17667. Cushioning material used to protect and immobilize the item within the container shall conform to Type III, Class B (1/2 inch thick) of Specification PPP-C-843. Place cushioned power unit in a close -fitting container conforming to Type I, Class 1 of Specification PPP-B-636. Closure of container shall be with 2-inch tape conforming to Class 2 of Specification UU-T-111. Apply one strip on the top and bottom seam, extend 3 inches on to each end panel. It shall then be placed in a close-fitting heat sealed bag fabricated from barrier material conforming to Class 1 of Specification MIL-B-131. Exhaust as much of the air from the bag as is practical.

5.1.1.5 Intermediate packaging.- Four power units, packaged as described in 5.1.1.4, shall be placed in a close-fitting intermediate container conforming to the requirements of Specification PPP-B-636, Type I, Class 2, Grade 6. The container shall be closed as specified in the appendix of the box specification using tape conforming to Specification PPP-T-76.

5.1.2 Level C.- Packaging shall be accomplished in a manner consistent with common carrier acceptance and to insure safe delivery to the first receiving activity.

5.2 Packing.- Packing shall be level A, B, or C as specified (See 6.1).

5.2.1 Level A.- The power units, packaged as described in 5.1.1.5, shall be packed in a wood-cleated plywood box, nailed wood box, wood-cleated fiberboard box or wirebound wood box conforming to the requirements of Specifications PPP-B-601, overseas type or PPP-B-621, Class 2 or PPP-B-591, overseas type or PPP-B-585, Class 3, respectively. The container shall be closed and strapped in accordance with the applicable container specification and/or the appendix thereto. The maximum gross weight of the shipping container shall not exceed 200 pounds by more than 10 percent.

5.2.2 Level B.- The power units, packaged as described in 5.1.1.5, shall be packed in a wood-cleated box, nailed wood box, wood-cleated fiberboard box or wirebound wood box conforming to the requirements of Specifications PPP-B-601, domestic type, or PPP-B-621, Class 1, or

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PPP-B-591, domestic type, or PPP-B-585, Class 1, respectively. The container shall be closed and strapped in accordance with the applicable container specification and/or the appendix thereto. The maximum gross weight of the shipping container shall not exceed 200 pounds by more than 10 percent.

5.2.3 Level C.- The power units, packaged as described in 5.1, shall be packed in conformance to common carrier acceptance at the lowest transportation rate available for the commodity and shall insure safe delivery to the first receiving activity.

5.3 Marking.- In addition to any special marking required by the contract or order, unit packages, intermediate packages and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packaging, packing and preservation.
(c) Availability of LSQAP's and Government inspection equipment (Sec. 4).
(d) Applicable stock number.
(e) Provisions for production model including date of delivery (See 3.1).

6.2 Size of packed container.- The size of the container will be determined by the type of packing and packaging of the power unit. The containers and bags shall be of a minimum size that will adequately contain the items.

6.3 Definitions.

6.3.1 Measuring equipment (M.E.) employs commercially available devices such as micrometers, calipers, height indicators, surface plates, sine bars, etc. This category also includes commercial test equipment such as meters, optical comparators, etc.

6.3.2 Special test equipment (STE) includes special inspection fixtures, instruments modified measuring equipment and special purpose commercially available equipment.

6.3.3 LSQAP (List of Supplemental Quality Assurance Provisions).- The SQAP is a contractual document which supplements Section 4 of a specification to provide quality assurance provisions for the requirements specified in Ordnance detail drawings.

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